



High Sensitivity Gold-Coated Photonic Crystal Fiber Sensor for Blood Component Detection

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Abstract

Photonic crystal fiber (PCF), with its periodic refractive index variations, enables unique light propagation properties such as single-mode guidance and high nonlinearity. This work focuses on the design and simulation of such a PCF, specifically optimized for identifying key blood components like white blood cells (WBC), red blood cells (RBC), plasma, water, and hemoglobin. A nanofilm gold layer is used as a plasmonic substance to enhance sensitivity, increase birefringence, optimize resolution, and maximize transmittance. An analysis and evaluation of the sensor efficacy of this suggested architecture is conducted using the finite element method (FEM). The analytes with high sensitivity to surface plasmon resonance (SPR) are injected into the outermost air holes where silica is serving as the base material. The analysis employed refractive index (RI) values of 1.33, 1.35, 1.36, 1.38, and 1.40 and a wavelength range of 1.1–2.1 μm . Key parameters, including birefringence index (BI), confinement loss, coupling length, power spectrum, transmittance, and power fraction, are meticulously calculated. The maximum birefringence, coupling length, transmittance, and power fraction are 2.17899×10^{-5} , 71059.43 μm , –28.8 dB, and 760.1, respectively. Additionally, the sensitivity analysis for wavelength provided significant insights, with the highest recorded wavelength sensitivity reaching 24,340.29 nm/RIU for X-polarization and 24,349.82 nm/RIU for Y-polarization at RI analyte 1.33. This work offers valuable contributions to the design and optimization of PCF-based blood component sensors.

Keywords PCF · SPR · Birefringence · Transmittance · Confinement loss

Introduction

Photonic crystal fiber (PCF) is a type of optical fiber with a unique periodic arrangement of air holes along its length, enabling exceptional control over light propagation [1, 2]. This structured design allows PCFs to achieve low transmission loss and high sensitivity, making them ideal for bio-sensing applications [3, 4]. Accurate detection and monitoring of blood components, such as white blood cells, red blood cells, plasma, water, and hemoglobin, are critical in medical diagnostics and patient health monitoring [5, 6].

Optical sensing, known for its label-free detection and real-time analysis capabilities, has emerged as a valuable tool in this domain [7–9]. PCFs stand out due to their ability to precisely manipulate light-matter interactions through tailored structural parameters, enabling the detection of various blood components with distinct refractive indices [10–15]. This versatility positions PCF-based sensors as a promising solution for high-precision biomedical diagnostics, providing rapid response times, high sensitivity, and compatibility with a wide range of wavelengths [16–21]. Moreover, PCF sensors offer low signal loss, structural adaptability, and potential for miniaturization, making them suitable for point-of-care testing and remote monitoring [22–25]. The ability of PCFs to create customized sensing platforms optimized for specific blood components further enhances their applicability in clinical and laboratory settings [8, 26–32]. This technology has the potential to revolutionize healthcare by enabling personalized medicine, real-time disease detection, and effective patient monitoring, thereby addressing key challenges in biomedical sensing [33–37].

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In 2019, Ying et al. developed an optical fiber sensing with split polarization with a pentagonal lattice design for rectification of refractive index within the near-IR range, achieving a resolution of 9.72×10^{-6} RIU and a wavelength sensitivity of 10,286 nm/RIU across an RI range of 1.32–1.38. Despite the sensor's high amplitude sensitivity of 544.0 RIU^{-1} and a figure of merit of 146.9, the overall sensitivities exhibited a very low response [3]. In 2019, Haofeng et al. studied an SPR fiber sensor with a multimode-no-core-multimode (MNM) structure, achieving a highest sensitivity of 11,792 nm/RIU and a refractive index of 2.04×10^{-5} RIU across the 1.3330–1.4102 RI range. Although the sensor reached the highest possible RI resolution, its sensitivity fell short of expectations [6]. In 2020, Veerpal et al. analyzed a D-shaped PCF biosensor using SPR and FEM, enhanced with indium tin oxide (ITO) and zinc oxide (ZnO) for detecting signals within the 1.30 to 1.38 range, achieving a wavelength sensitivity of 10,000 nm/RIU and a refractive index resolution of 2.0×10^{-5} . Although the sensor is suitable for medical diagnostics and environmental monitoring, its sensitivity was unsatisfactory [4]. In the same year (2020), Nan et al. developed a mid-IR D-shaped sensor based on SPR, achieving a wavelength sensitivity of 11,500 nm/RIU, a refractive index resolution of 8.7×10^{-6} RIU, and a figure of merit of 142. Despite its potential for environmental monitoring, the sensor's sensitivity response was low [5]. In 2022, Abdullah et al. created a dual polarized SPR sensor that can detect RI values between 1.33 and 1.39 by employing a decentralized propagation-controlled core PCF. The sensor attained peak wavelength sensitivity of 13,700 nm/RIU and 15,400 nm/RIU for X- and Y-polarization respectively. Despite the slightly low sensitivities, the sensor demonstrated a good figure of merit [1]. In 2023, PCF-SPR sensing containing a u-grooved specific layer was created by Abdullah et al. for bio sensing and distant use. It covered a range of 1.29–1.40 RI and had a highest amplitude sensitivity of 1189 RIU^{-1} , a wavelength sensitivity of 12,500 nm/RIU, and a resolution of 8.0×10^{-6} RIU. The device's sensor is appropriate for point-of-care bio-sample screening due to its remarkable responsiveness. Despite its high resolution, the maximum amplitude and wavelength sensitivities were modest [2].

The study presents a method for designing and simulating a circular core PCF specifically for blood component sensing, assessing its ability to detect white blood cells, red blood cells, plasma, water, and hemoglobin. For the proposed sensor simulations, FEM method is applied as the mode solver. This method is employed to solve Maxwell's equations within the proposed PCF structure. A mesh with a total of 14,478 elements was used in the simulations, consisting of 1242 boundary elements and 108 vertex elements. A minimum element quality of 0.4191 was attained, signifying that the mesh sufficiently captured the geometric

details required for precise mode solution. A convergence analysis is performed in order to confirm that the mode solutions are stable. Using amorphous as the substrate material, strongly SPR responsive RI analytical compounds are inserted into the cladding's furthest air pores. The analysis reveals an uppermost wavelength sensitivity of 24,340.29 nm/RIU for X-polarization and 24,349.82 nm/RIU for Y-polarization, with a thorough examination of parameters and also achieving its maximum birefringence, coupling length, transmittance, and power fraction are 2.17899×10^{-5} , 71,059.43 μm , -28.8 dB , and 760.1.

Structure Design and Methodology

The research focuses on a circular core configuration for the creation of the PCF. Figure 1 depicts the cross-sectional area of the suggested PCF, which is composed of a regular arrangement of air holes within a silica matrix, creating the photonic crystal cladding.

Sensing materials consist of a superimposed thin layer of gold. The 50-nm gold layer is indicated by the color red. Within the single layer sensing substances, dual layer air holes in circles with matrix pitches of $\Lambda_1 = 3.66 \mu\text{m}$ and $\Lambda_2 = 2.12 \mu\text{m}$ are put together in a cycloidal structure. The gold (Au) coating functions as a plasmonic layer in the sensor by supporting SPR. When light interacts with the active plasmonic material gold layer, it excites collective oscillations of electrons at the surface, resulting in enhanced electromagnetic fields. This amplification increases the sensor's sensitivity to changes in the refractive index (RI) of analytes. As the RI of the surrounding medium changes, it affects the resonance conditions, leading to shifts in the wavelength of the reflected light. By monitoring these shifts, the sensor can accurately detect and quantify different blood components based on their unique RI values. Here, gold is preferred as the plasmonic material in many SPR sensors due to its strong SPR effect, chemical stability, and ease of functionalization, making it ideal for biosensing applications [38–42]. The analytes are introduced in the cycloidal arrangement at a lattice pitch of $\Lambda_3 = 6.86 \mu\text{m}$, outside the sensing materials. The diameter of the air holes and analytes holes is defined as $D_1 = 1.25 \mu\text{m}$ and $D_2 = 1.75 \mu\text{m}$. Regarding the construction, the cladding layer's diameter is $10 \mu\text{m}$.

To get the best possible sensitivity and performance, the suggested gold-coated photonic PCF sensor is made by going through a number of crucial procedures. First, silica glass is chosen as the main material for the sensor because of its wide range of transparency and low loss qualities, which make it perfect for optical applications. Subsequently, the silica matrix is etched to create air holes. A nano-film gold layer is applied using physical vapor deposition (PVD) [43], enhancing sensitivity through SPR. Lastly, biological

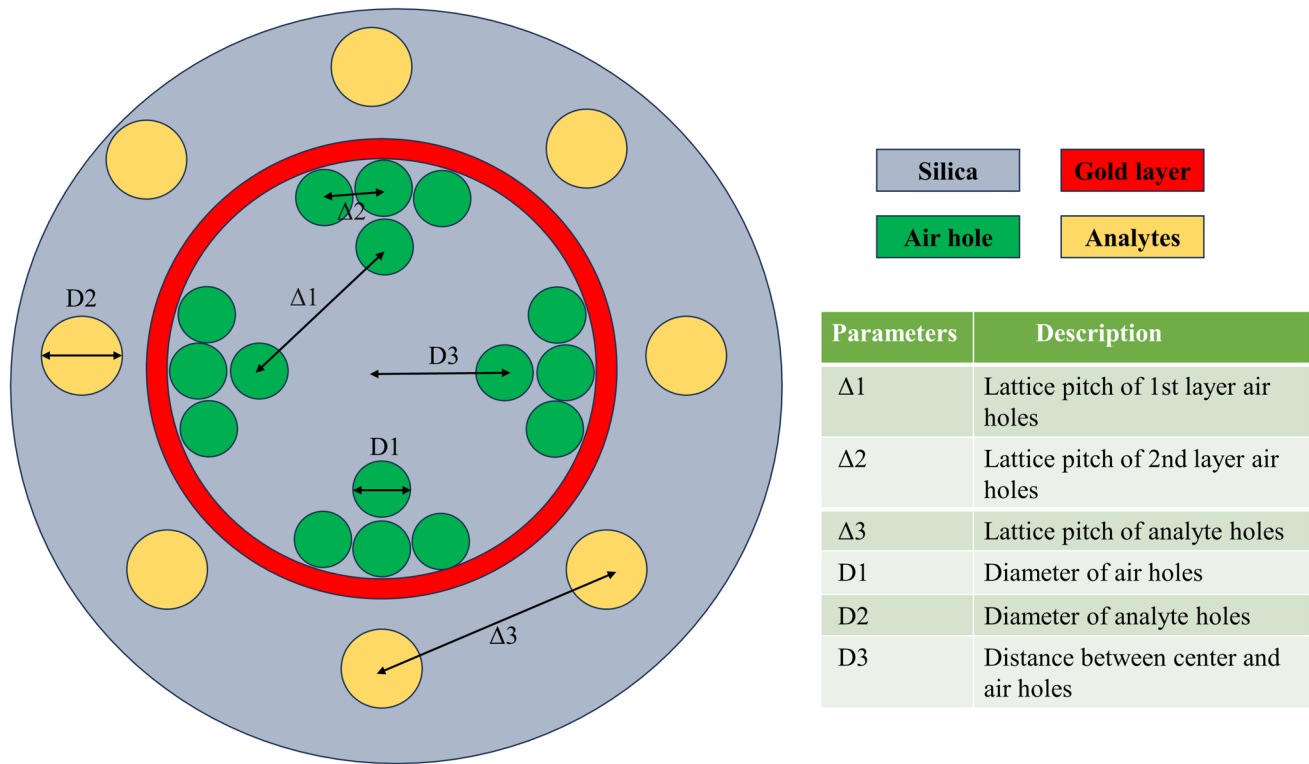


Fig. 1 The cross-sectional region of the suggested PCF sensor

analytes are injected into the air holes for detection, including plasma, red blood cells (RBC), and white blood cells (WBC). Using an optical setup, the sensitivity, transmittance, and birefringence of the sensor are measured to evaluate its performance. For high precision and application in biological sensing, this meticulous production procedure is essential.

The choice of silica as the primary material for the PCF is based on its advantageous characteristics, including low loss and great transparency across a broad spectrum of wavelengths. Silica glass (amorphous) is utilized as the substrate material, and its RI can be determined by employing the Sellmeier equation [27]:

$$n(\lambda) = \frac{P_1 \lambda^2}{\lambda^2 - Q_1} + \frac{P_2 \lambda^2}{\lambda^2 - Q_2} + \frac{P_3 \lambda^2}{\lambda^2 - Q_3} \quad (1)$$

Here, n represents the effective refractive index of the fused silica, whereas λ represents the wavelength in micrometers. The Sellmeier constants consist of six parameters: P_1 , P_2 , P_3 , Q_1 , Q_2 , and Q_3 [27]. The simulations used the finite element technique to solve the equations proposed by Maxwell in the PCF structure. The FEM allows for precise modeling of complex geometries and material distributions. The proposed structure has the advantage of enhancing sensitivity through the nanofilm gold layer, which optimizes

light-matter interactions, thereby improving the sensor's resolution and transmittance. Its dual-layer cycloidal design allows for precise analyte detection with high wavelength sensitivity and minimal confinement loss, making it suitable for accurate identification of blood components [37]. The intricate design of the photonic crystal fiber (PCF), involving precise arrangements of air holes and the application of a thin gold layer, requires advanced fabrication techniques. This complexity can increase production time and costs.

Figure 2 provides a detailed illustration of the experimental setup designed to evaluate the performance of the proposed circular core PCF for blood component detection. This setup is meticulously constructed to ensure precise measurement of the sensor's response to various refractive indices that correspond to different blood components. The optical spectrum analyzer (OSA), optical tunable source (OTS), and polarization controller were part of this configuration. Each of these three components is connected via the single mode (SM) fiber. The pump spectrum can exert control on both the outlet (analyte) and the inlet (analyte). The laser source is capable of emitting light within the wavelength range of 1.1–to 2.1 μm , which is then coupled into the PCF. The transmitted light is analyzed by the OSA to measure changes in both intensity and wavelength. The refractive index calibration unit allows for precise adjustment of refractive index values between

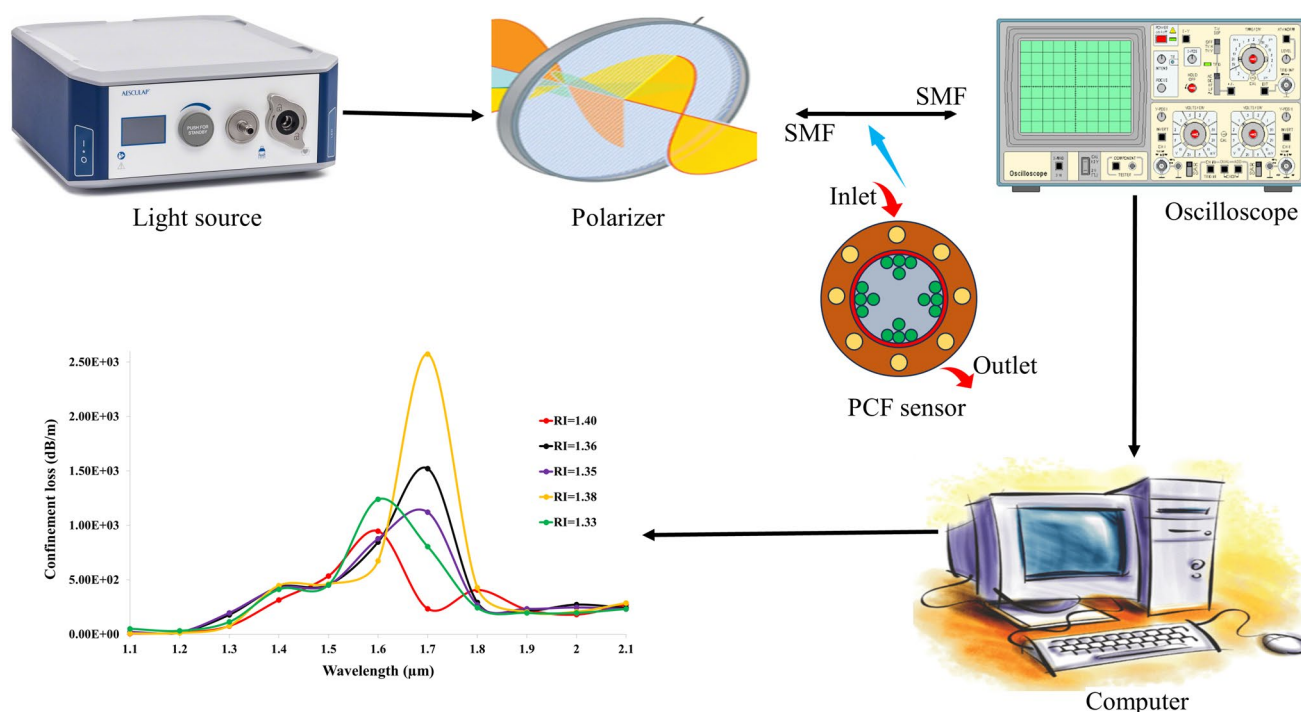


Fig. 2 Experimental setup of the proposed PCF sensor

1.33, 1.35, 1.36, 1.38, and 1.4, accurately simulating the refractive indices of various blood components. The sensor's ability to detect analytes with a wide refractive index (RI) range of 1.1–2.1 μm is significant because it enhances its versatility in biomedical diagnostics. This broad RI range allows the sensor to accurately identify and analyze various blood components, such as white blood cells, red blood cells, plasma, and hemoglobin, each of which exhibits different RI values. This capability facilitates real-time, label-free analysis, making the sensor highly applicable for point-of-care testing and other medical applications. Data collected from the OSA is analyzed to determine the sensitivity and specificity of the PCF by calculating key metrics such as birefringence, confinement loss, and sensitivity. This experimental setup validates the PCF's design, demonstrating its potential for real-time, label-free blood analysis. The proposed PCF structure, featuring a circular core and gold coating, is specifically designed to enhance sensitivity by leveraging SPR. Key to its application is the introduction of sensing materials into the cyclidal arrangement of air holes in the fiber's cladding. The sensor's architecture allows for real-time, label-free blood analysis by improving optical sensitivity, which is critical for biomedical diagnostics. The gold nanofilm amplifies light interaction, significantly improving the transmittance and power spectrum of the sensor, making it more efficient in medical and diagnostic applications. In short, the PCF structure enhances performance in blood component

detection by optimizing sensitivity and transmittance, enabling precise and reliable analysis in real-time diagnostics.

Result Analysis

Figure 3 provides a comprehensive depiction of the mode field distribution for the sensor that has been proposed. The figure illustrates 07 distinct modes: (a) the mode of X-polarization; (b) the mode for Y-polarization; (c) 1st surface plasmon polariton mode; (d) 2nd surface plasmon polariton mode; (e) 4th surface plasmon polariton mode; (f) 6th surface plasmon polariton mode; and (g) 8th surface plasmon polariton mode. Each mode corresponds to a distinct arrangement of the electromagnetic field distribution within the sensor, which presumably indicates the sensor's response to various polarization states and the stimulation of SPP mode. These modes are crucial for comprehending the sensor's performance, especially in applications that involve polarization and plasmonic effects.

Birefringence

Birefringence is an aberration that is officially recognized as the double refraction of light in a material that is molecularly organized and transparent. This occurrence is displayed by the presence of changes in refractive index that are depending on the orientation of the material.

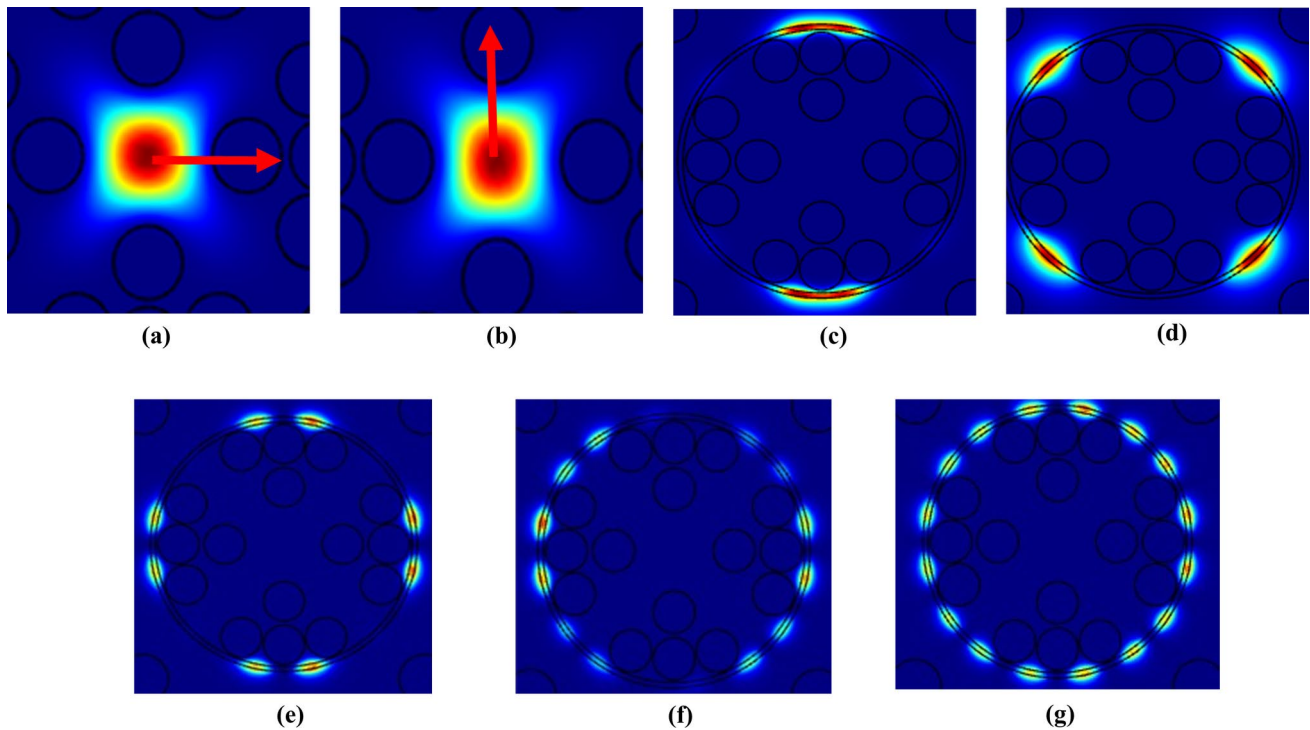


Fig. 3 Mode field distribution of **a** X polarization, **b** Y polarization, **c** 1st SPP mode, **d** 2nd SPP mode, **e** 4th SPP mode, **f** 6th SPP mode and **g** 8th SPP mode of the proposed sensor

The variation between the maximal modes on the X- and Y-poles determines the birefringence (B_i), sometimes known as the effective index variation. The variation is contingent upon the accessible wavelength. The equation of birefringence that follows is used to determine its value [8].

$$B_i = |n_e^i - n_o^i| \quad (2)$$

where i denotes either the Y- or the X-polarization for the suggested structural configuration, the refractive indices that are experienced are denoted by n_e and n_o . With regard to wavelength, the birefringence of X- and Y-polarity for the analytes 1.33, 1.35, 1.36, 1.38, and 1.40 is shown in Fig. 4. These analytes belong to the group of analytes. In the process of increasing the wavelength from 1.1 to 2.1 μm , the B_i curves exhibit a gradual and continual increase. The RI disparities between the super modes are growing as the wavelength grows, which is the reason why the curves are spreading out across the wavelength. There exists an additional correlation between the analyte as well as the B_i , whereby larger analytes exhibit more B_i properties, while smaller analytes exhibit lesser B_i factors. As a consequence of this, the B_i characteristics of 1.79×10^{-5} is obtained for the largest RI analyte 1.40, and the B_i characteristics of 7.74×10^{-6} is obtained for the lowest RI analyte 1.33.

Coupling Length

The coupling length pertains to the spatial extent across which power is exchanged between two modes in a waveguide, such as a PCF. The parameter plays a crucial role in establishing the distance over which light can connect between different modes inside the fiber. B_i influences the coupling length by dictating the manner in which light interacts with the fiber's structure. Greater B_i can cause shorter coupling lengths as a result of increased interaction between the modes, whereas lower B_i can result in longer coupling lengths. The sensitivity and efficiency of the sensor are determined by the coupling length.

By modifying the coupling length, the sensor can be tuned to identify precise wavelengths or alterations in the nearby surroundings, such as changes in temperature, pressure, or the existence of specific compounds. The sense efficiency of the PCF sensor may be assessed by the computation of coupling length, which is the primary independent variable. This parameter can be easily determined by using the following equation, after computing the B_i . The equation can be rewritten as follows [25]:

$$L_c(\mu\text{m}) = \frac{\lambda}{2 * B_i} \quad (3)$$

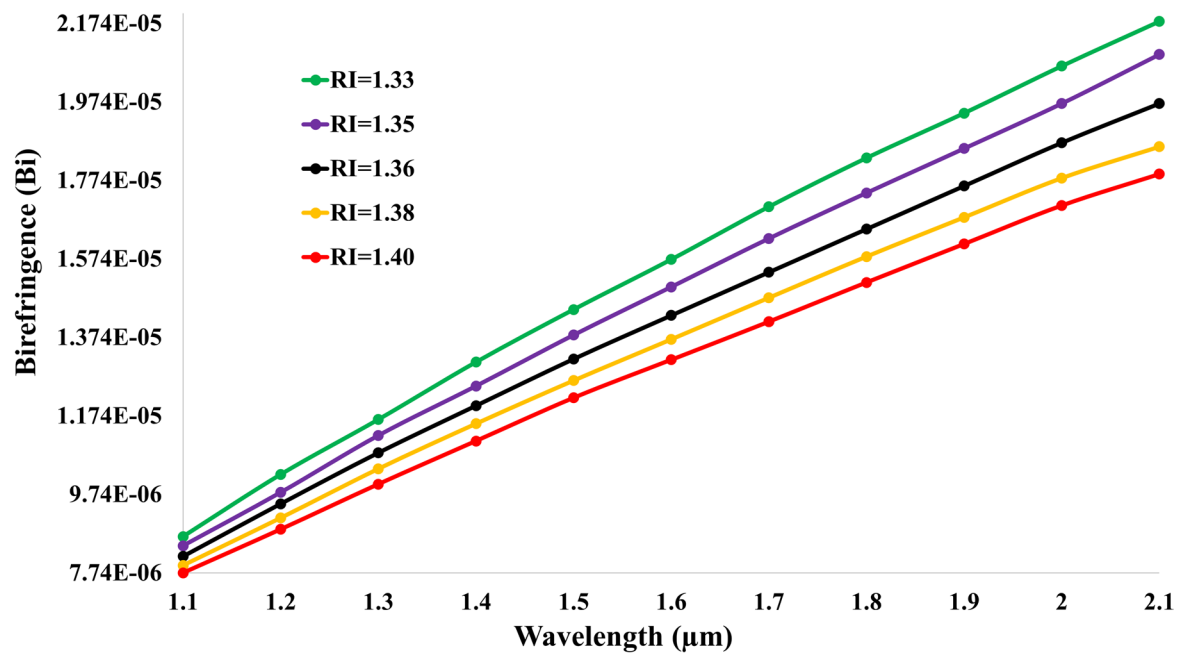


Fig. 4 The wavelength-dependent birefringence spectrum for different analytes

The variables in the equation are as follows: L_c represents the coupling length, λ represents the functional wavelength in micrometers (μm), and B_i represents birefringence. Figure 5 depicts the correlation between the coupling length versus the wavelength for the proposed PCF sensor. The

graph illustrates the relationship between wavelength and coupling length, depicting several curves corresponding to different modes or conditions within the sensor. As the wavelength increases from 1.1 to 2.1 μm , the coupling length consistently decreases across all displayed modes.

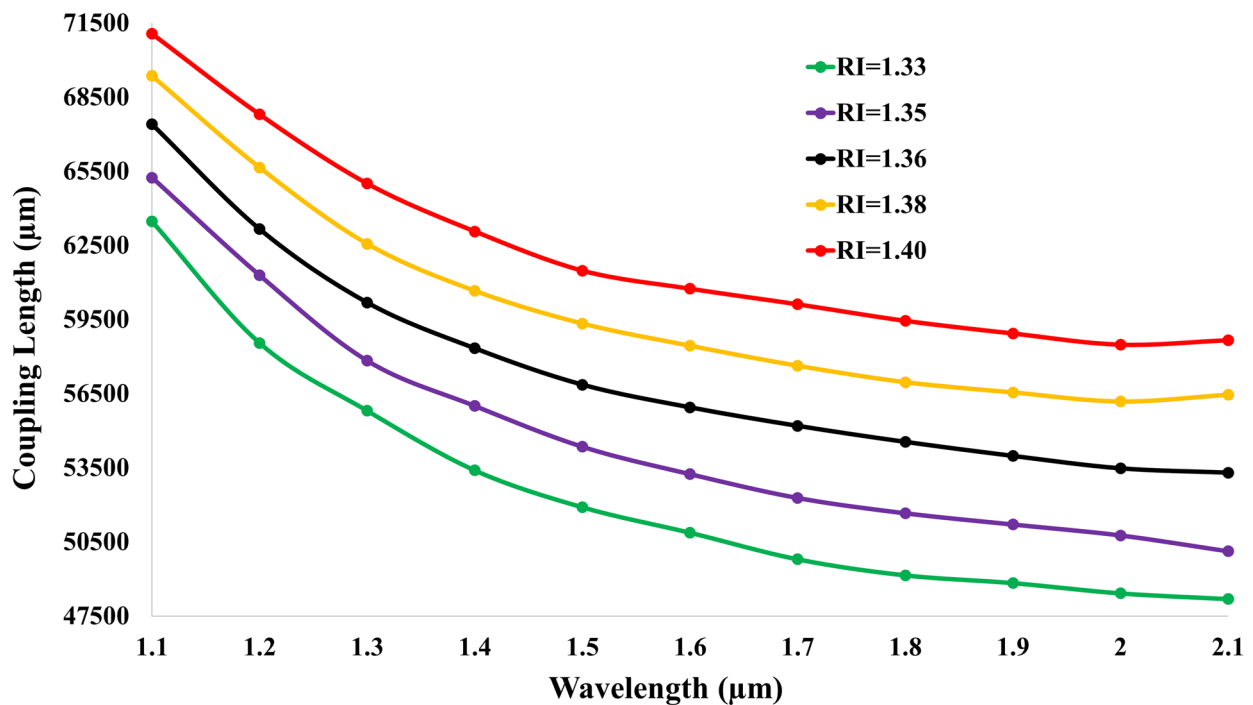


Fig. 5 The coupling length spectrum of the proposed sensor in relation to the wavelength for different analytes

The curves indicate that at shorter wavelengths, the coupling length is greater, with a peak value occurring at 1.1 μm , which suggests a weaker interaction and a longer distance required for power transfer between modes. Conversely, as the wavelength approaches 2.0 μm , the coupling length reaches its minimum, indicating a stronger interaction and shorter distances for power transfer. Each curve follows a similar trend of decreasing coupling length with increasing wavelength, though at varying levels, likely reflecting different polarization modes or specific conditions within the PCF sensor.

Power Spectrum

The power spectrum represents how the power (or intensity) of a signal is distributed across various wavelengths or frequencies. Figure 6 displays the output power spectrum as a function of wavelength and analytes, ranging from approximately 1.1 to 2.1 μm and 1.33, 1.35, 1.36, 1.38, 1.40 respectively. Each colored curve corresponds to the power spectrum for different conditions or configurations, showing how the power varies across the wavelength range. The power spectrum has multiple peaks and valleys, indicating varying levels of intensity at different wavelengths. A periodic rhythm is observed in the transmission power spectrum, which is dependent on the birefringence and coupling length. The following formula may be used to figure out it, and it makes it evident that the curves' intensities must remain in the range of 0 to 1 [28]:

$$P_{\text{out}}(w) = \frac{B_i * \pi * l}{\lambda} \quad (4)$$

where, birefringence is denoted by B_i , wavelength is denoted by λ , and l is the operative fiber length. The spectrum of 1.38 is lagged most and 1.40 is forward most.

Transmittance

Upon computing the power that is produced spectrum, it becomes straightforward to determine the transmittance spectrum by employing the below formula [20]:

$$T_s(\text{dB}) = 10 \log_{10} \frac{P_{\text{out}}}{P_{\text{in}}} \quad (5)$$

Here, the transmittance is denoted by T_s , P_{out} is highest output, and P_{in} is lowest input.

Figure 7 shows the transmittance behavior of the designed PCF across different wavelengths. In the study, the transmittance output is crucial for understanding how efficiently light propagates through the PCF when it interacts with different blood components, such as WBC, RBC, plasma, water, and hemoglobin. The graph covers a wavelength range from approximately 1.1 to 2.1 μm , consistent with the wavelength range. It also displays multiple curves, each corresponding to a different refractive index 1.33, 1.35, 1.36, 1.38, and 1.40. The magnitudes of the transmittance graphs range from 0 dB to -29.80 dB. Higher analytes have a greater number of downward peaks, whereas lower analytes exhibit a smaller

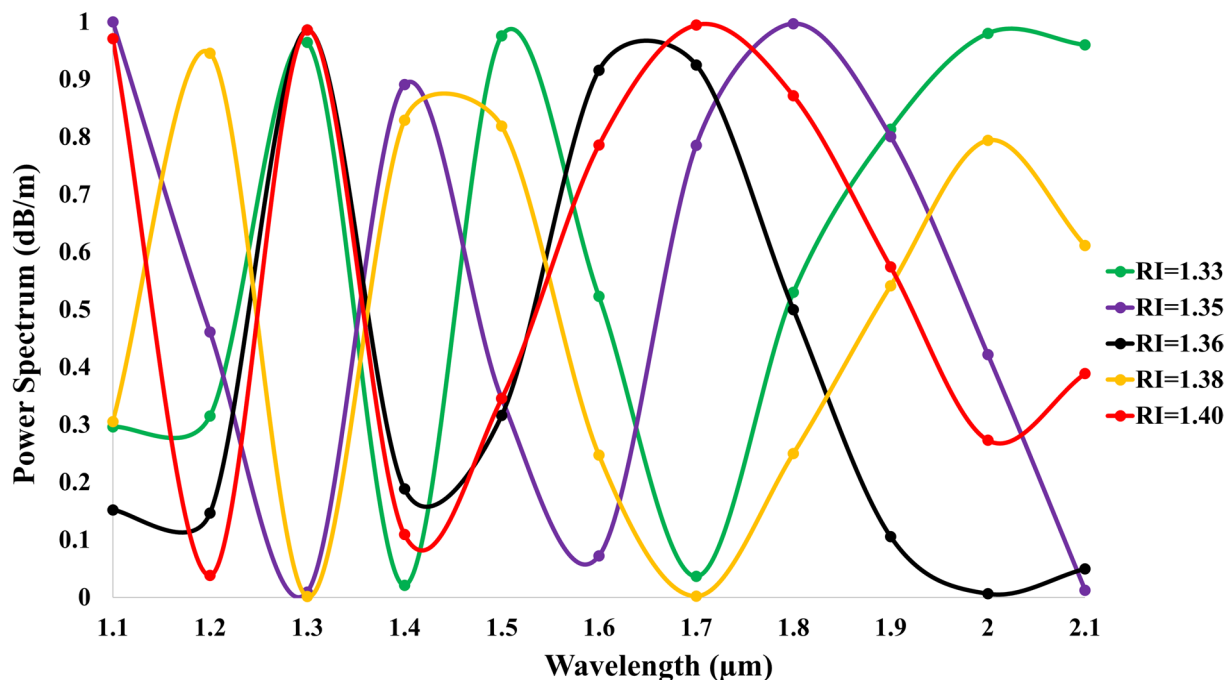


Fig. 6 The power spectrum of the proposed sensor with respect to wavelength

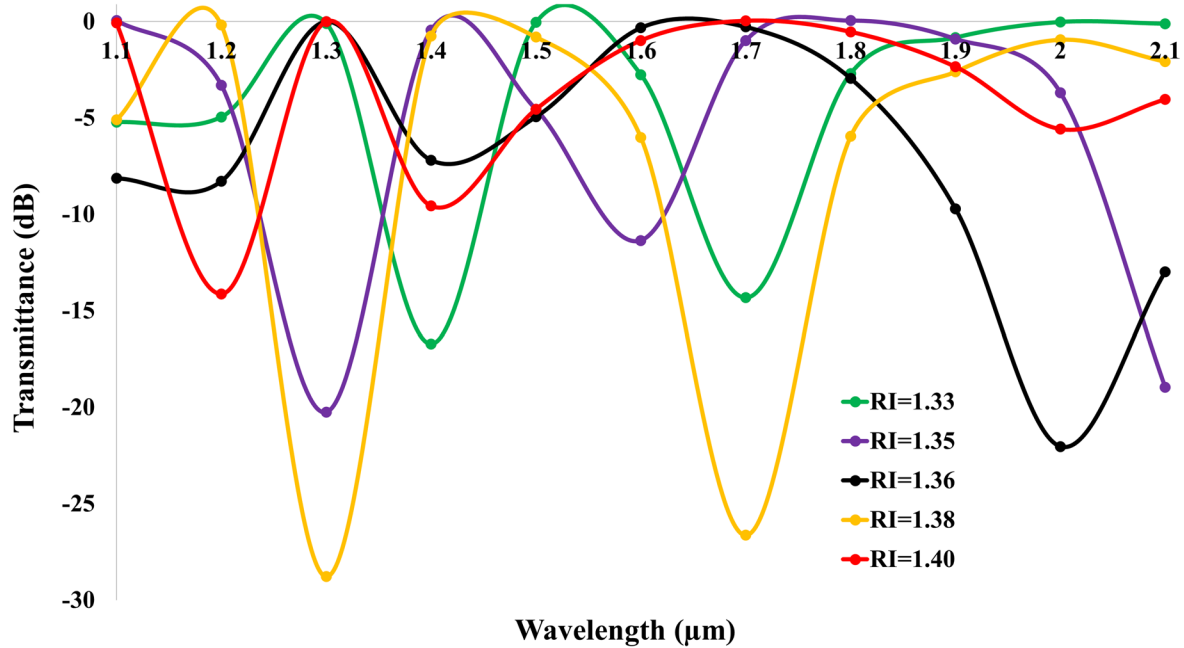


Fig. 7 The transmittance spectrum of the proposed sensor with respect to wavelength

number of downward peaks. The transmittance profile for the analytes 1.33, 1.35, 1.36, 1.38, and 1.40 correspondingly is as follows: -16.17 dB, -20.25 dB, -22.1 dB, -28.8 dB, and -14.13 dB. The transmittance waveform has a maximum value of 1.38 in the direction that is forward, while a minimum value of 1.4 is seen with a few strong downward peaks in the backward direction. The analytes with values of 1.33, 1.35, 1.36, and 1.40 exhibit three backward peaks, whereas the analyte with a value of 1.38 shows two backward peaks. By analyzing these transmittance curves, it is possible to determine which wavelengths are most effective for detecting specific blood components.

Power Fraction

Figure 8 illustrates the power fraction as a function of wavelength for the designed PCF, highlighting its efficiency and sensitivity in detecting blood components. The power fraction of the fiber is calculated through Eq. (6) [6]:

$$P_f = \frac{P_{\text{input}}}{P_{\text{output}}} \quad (6)$$

where, P_{input} and P_{output} refer to the power input and output, respectively. Notably, three significant peaks are observed at wavelengths around $1.3 \mu\text{m}$, $1.7 \mu\text{m}$, and $1.9 \mu\text{m}$. The highest peak, occurring at $1.3 \mu\text{m}$ with a power fraction of approximately 800, indicates a strong optical interaction at this wavelength, suggesting high sensitivity for specific blood components. The second peak at $1.7 \mu\text{m}$, with a power

fraction around 550, also points to substantial sensitivity, while the $1.9\text{-}\mu\text{m}$ peak, though lower at around 100, still represents a noteworthy interaction. The yellow curve dominates these peaks, particularly at $1.3 \mu\text{m}$ and $1.7 \mu\text{m}$, demonstrating sharp rises and falls, indicating strong interactions at these specific wavelengths. Other curves exhibit much smaller power fractions across the wavelength range, suggesting weaker interactions. The behavior of the power fraction across wavelengths suggests that $1.3 \mu\text{m}$, $1.7 \mu\text{m}$, and $1.9 \mu\text{m}$ are optimal wavelengths for blood component sensing due to the strong optical power guidance observed.

Confinement Loss

The confinement loss spectrum is important for PCF sensor sensing efficiency. The significance of this number, which is directly related to the actual component of the effective index, cannot be overstated and may be calculated as follows [2]:

$$\alpha_{\text{loss}} = 8.686 \times \frac{2\pi}{\lambda} \times \text{Im}(n_e) \times 10^6 \quad (7)$$

$\text{Im}(n_e)$ is the imaginary portion of the effective index and λ is the wavelength. Figure 9 displays the confinement loss of the PCF across wavelengths for X-polarization. Starting at $1.1 \mu\text{m}$, confinement loss is minimal, close to zero, and remains low until around $1.3 \mu\text{m}$. As the wavelength increases, the loss gradually rises, with a significant peak at $1.7 \mu\text{m}$, where the yellow curve reaches its maximum value, exceeding 2500. This peak indicates a strong interaction

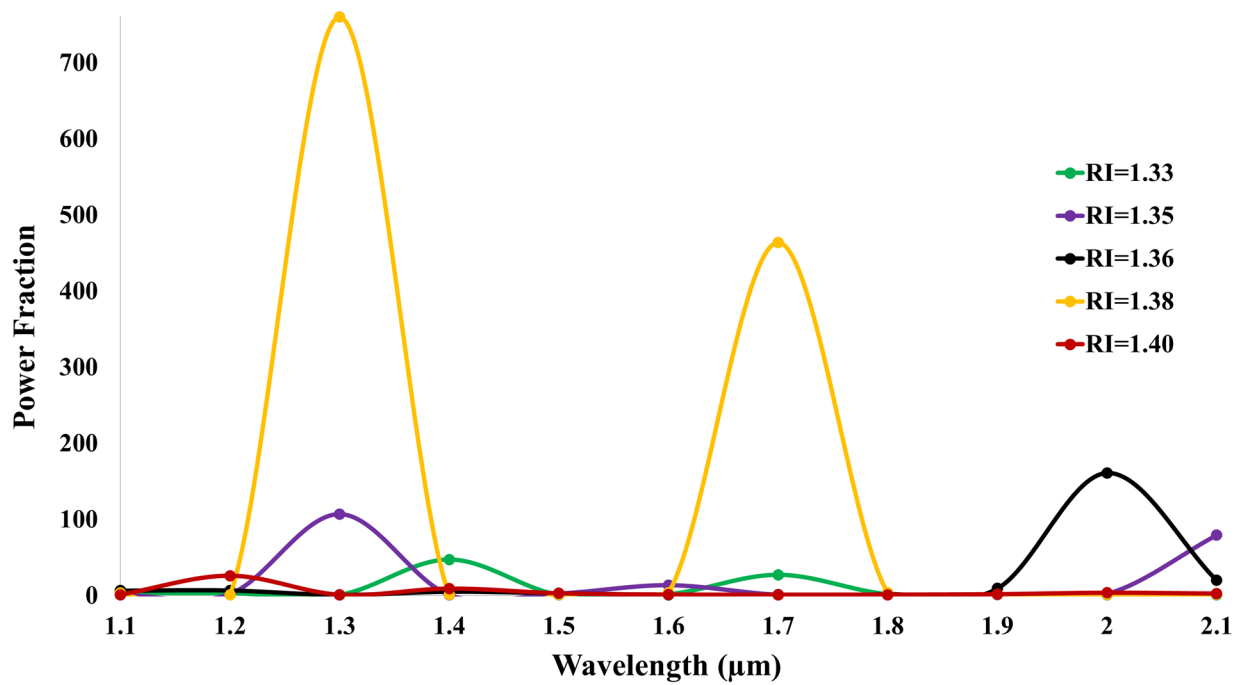


Fig. 8 The power fraction spectrum of the proposed sensor with respect to wavelength

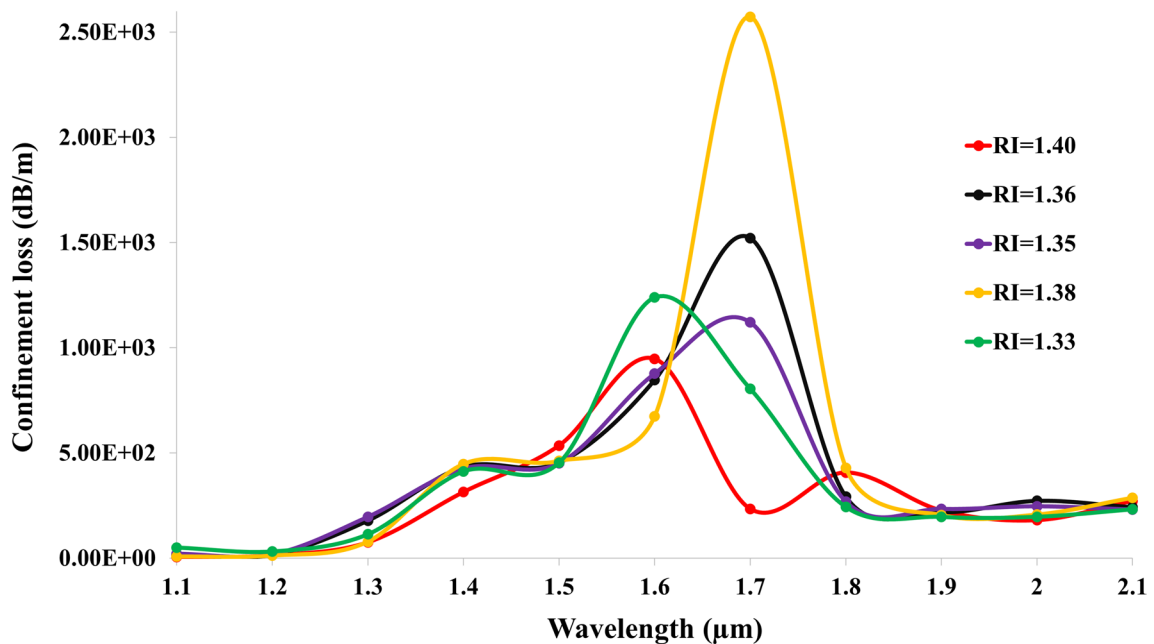


Fig. 9 The confinement loss spectrum of the proposed sensor with respect to wavelength for X-polarization

between light and the fiber's structure, leading to higher loss at this specific wavelength. After 1.7 μm , the loss sharply decreases, particularly in the purple curve, which drops back down to lower levels by 1.8 μm . The other curves also show a decline in loss, though at different rates. By 2.1 μm , the confinement loss stabilizes, returning to values similar to

those observed at the starting point. This suggests that the PCF experiences the highest loss near 1.7 μm , with lower losses at both shorter and longer wavelengths. The sharp rise and fall in confinement loss around 1.7 μm highlight the wavelength's critical influence on the fiber's optical performance. Figure 10 shows the confinement loss of the

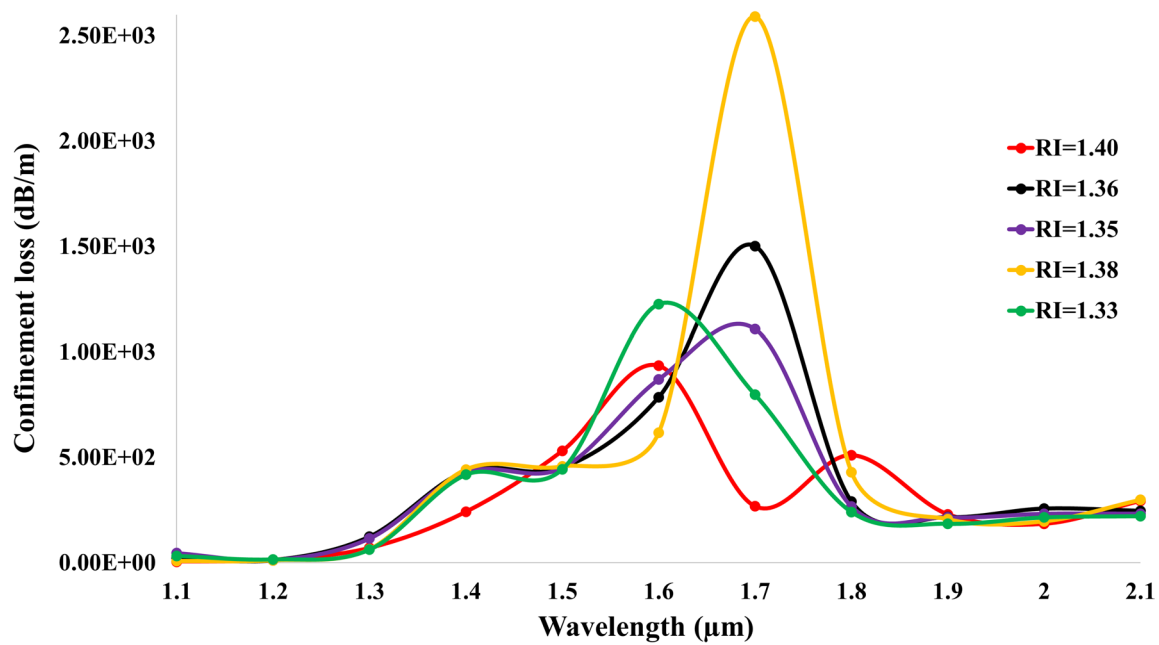


Fig. 10 The confinement loss spectrum of the proposed sensor with respect to wavelength for Y-polarization

PCF across wavelengths for the Y-polarization. The graph begins at a wavelength of 1.1 μm , where the confinement loss is minimal, close to zero, and remains low until around 1.3 μm . As the wavelength increases, the loss gradually rises, with significant peaks occurring around 1.7 μm . The yellow curve reaches its maximum value, with a confinement loss exceeding 2500, indicating a strong interaction between light and the fiber's structure, leading to higher loss at this specific wavelength. The black, green, cyan, and purple curves also peak, with losses reaching approximately 2000, 1500, 1200, and 1000, respectively. After 1.7 μm , the loss sharply decreases, particularly in the red curve, which drops back down to lower levels by 1.8 μm . The other curves also show a decline in loss, though at different rates. By 2.1 μm , the confinement loss stabilizes, returning to values similar to those observed at the starting point. This pattern suggests that the highest loss in the fiber occurs near 1.7 μm , with lower losses at both shorter and longer wavelengths, highlighting the critical influence of this wavelength on the fiber's optical performance. Here, the loss curves are not maintaining a regular shift with increasing RI due to several contributing factors related to the interaction of light with the PCF structure and the SPR mechanism. The interaction between the guided mode and the surface plasmon mode is highly sensitive to the refractive index, and as the RI increases, the resonance conditions shift nonlinearly. This is evident in the irregular progression of loss curves at different RI values (1.33, 1.35, 1.36, 1.38, 1.40). It is also influenced by the dispersion of the guided modes in the fiber, which affects how the resonance wavelength shifts as the RI

changes. The refractive index variations are not linear, which leads to different confinement losses and nonuniform shifting across the wavelength range of 1.1–2.1 μm .

Sensitivity Response

A PCF sensor's sensing capability may be assessed using wavelength or amplitude probing methods. The following formula calculates wavelength probe sensitivity response [29]:

$$S(\lambda) = \frac{\Delta\lambda_{\text{peak}}}{\Delta n_r} \quad (8)$$

where, $\Delta\lambda_{\text{peak}}$ is the disparity in wavelength alterations between the peaks and Δn_r is the variation of RI analytes. The wavelength sensitivity responses to RI analytes 1.33, 1.35, 1.36, 1.38, and 1.40 for X-polarization are 24,340.28708 nm/RIU, 24,161.8932 nm/RIU, 24,171.14226 nm/RIU, 24,003.94307 nm/RIU, and 22,847.1369 nm/RIU, respectively. Therefore, the topmost wavelength sensitivity reaction of 24,340.28708 nm/RIU is observed for X-polarity at the analyte value 1.33. Again, for Y pole the wavelength sensitivity responses are 24,349.81656 nm/RIU, 24,170.20114 nm/RIU, 24,179.49186 nm/RIU, 23,954.38729 nm/RIU, and 22,762.58831 nm/RIU, respectively for the analytes. Therefore, for Y-polarity the topmost wavelength sensitivity reaction is 24,349.81656 nm/RIU at analyte 1.33. A comparison of the similarities and differences between the PCF sensors that we have proposed and those that have been published

Table 1 Comparison table of proposed PCF sensors vs. previously published PCF sensors

Structure name	Wavelength sensitivity (nm/RsU)	Year	Reference
SPR based PCF sensor with ITO layer	6701.03	2022	[29]
A plasmon sensor utilizing immobilized gold and titanium dioxide	12400	2022	[30]
A novel B_7 -core photonic crystal fiber	6930.00	2022	[32]
Two novel D -shaped SPR PCFs with dual-core	12,320.10	2019	[33]
Dual plasmonic layered LSPR-PCF sensor	11,088.50	2023	[34]
Hemoglobin on external gold-coated PCF	4500.00	2022	[35]
Gold-coated PCF for blood component analysis	24,340.29	2024	This work

previously is presented in Table 1. Our proposed PCF sensor performs far better than other PCF sensors that have been published previously, as shown in Table 1, which offers an unambiguous proof of this.

The advantage of the PCF structure over other D -shaped SPR sensors lies in its ability to offer higher sensitivity and better control over light propagation due to its unique geometry. This results in enhanced detection of blood components through improved birefringence, transmittance, and wavelength sensitivity. The proposed design achieved a peak sensitivity of 24,340.29 nm/RIU, which is significantly higher compared to previously published D -shaped SPR sensors [38–42].

Conclusion

Although the design and simulation of the circular core photonic crystal fiber presented in this study demonstrates its significant potential for blood component sensing. The FEM method is employed to calculate various parameters, with a nanofilm gold layer used as the active plasmonic material and silica serving as the base material. The careful analysis of key parameters such as birefringence index, confinement loss, and coupling length, along with the evaluation of sensitivity across a refractive index of 1.33, 1.35, 1.36, 1.38, and 1.40 and wavelength range of 1.1–2.1 μm , has yielded promising results. The highest recorded wavelength sensitivity is 24,349.82 nm/RIU. These findings confirm the effectiveness of the proposed PCF design in accurately identifying blood components, offering a robust approach for future biomedical sensing applications. The research provides critical insights into the optimization of PCF-based sensors, advancing the field of optical fiber sensing in medical diagnostics.

Author Contributions Conceptualization, Hasan Abdullah, Sonia Akter; methodology, Sonia Akter, Hasan Abdullah; software, Sonia Akter; validation, Hasan Abdullah; formal analysis, Sonia Akter; investigation, Sonia Akter; resources, Hasan Abdullah; data curation, Sonia Akter; writing—original draft preparation, Sonia Akter;

writing—review and editing, Hasan Abdullah; visualization, Hasan Abdullah; supervision, Hasan Abdullah.

Data Availability Not applicable.

Declarations

Ethics Approval This research is not directly related to humans or animals.

Competing Interests The authors declare no competing interests.

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